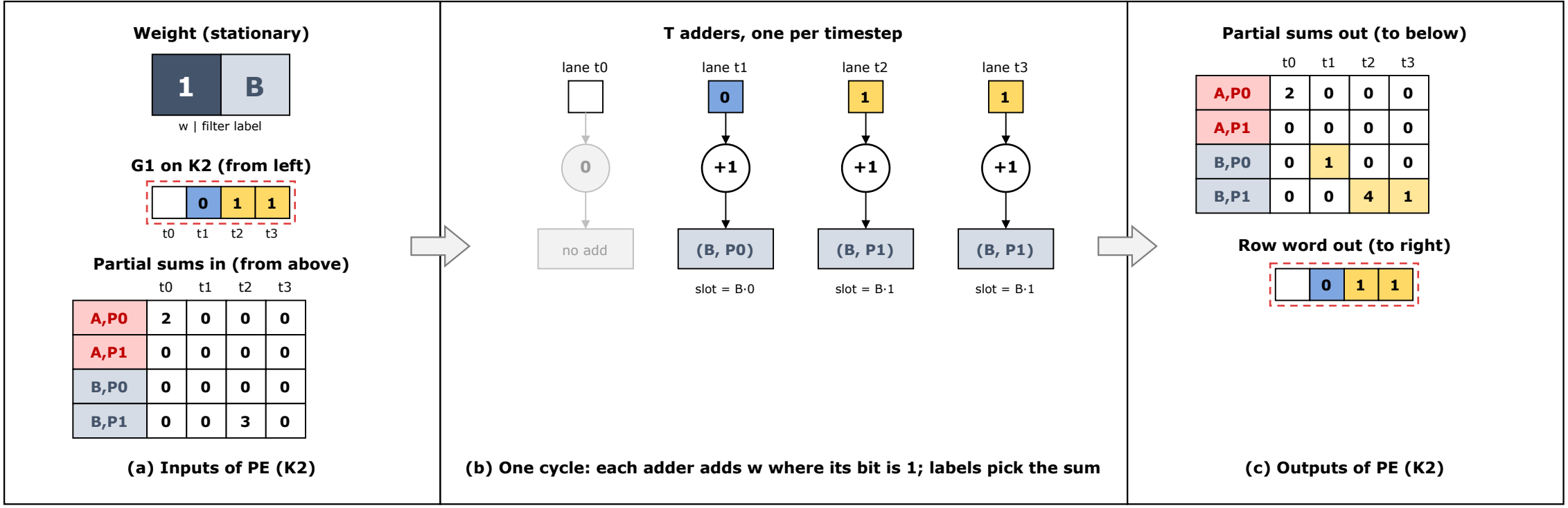
Weight (stationary)
1
B
w | filter label
G1 on K2 (from left)
0
1
1
t0 t1 t2 t3
Partial sums in (from above)
t0 t1 t2 t3
A,P0 2 0 0 0
A,P1 0 0 0 0
B,P0 0 0 0 0
B,P1 0 0 3 0
(a) Inputs of PE (K2)
T adders, one per timestep
lane t0
lane t1
lane t2
lane t3
0
no add
0
1
1
+1
+1
+1
(B, P0)
(B, P1)
(B, P1)
slot = B·0
slot = B·1
slot = B·1
(b) One cycle: each adder adds w where its bit is 1; labels pick the sum
Partial sums out (to below)
t0 t1 t2 t3
A,P0 2 0 0 0
A,P1 0 0 0 0
B,P0 0 1 0 0
B,P1 0 0 4 1
Row word out (to right)
0
1
1
(c) Outputs of PE (K2)
spike of P0
spike of P1
spike of P2
spike of P3
group
1
updated